Customer Shopping Behavior Analysis

Python | SQL | Power BI

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1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, discount usage, and subscription behavior to guide strategic business decisions.

The work follows an end-to-end workflow: data cleaning and exploratory analysis in **Python**, 32 business questions answered with **SQL**, and an interactive **Power BI** dashboard to present the findings.

2. Dataset Summary

- **Rows:** 3,900 (one record per purchase, 3,900 unique customers)
- **Columns:** 18 in the raw data; 19 after cleaning and feature engineering
- **Customer demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method
- **Missing data:** 37 values in the Review Rating column (no duplicate records found)
- **Tools used:** Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, Power BI

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

feature	count	mean	std	min	25%	50%	75%	max
age	3900	44.07	15.21	18.0	31.0	44.0	57.0	70.0
purchase_amount	3900	59.76	23.69	20.0	39.0	60.0	81.0	100.0
review_rating	3900	3.75	0.71	2.5	3.1	3.8	4.4	5.0
previous_purchases	3900	25.35	14.45	1.0	13.0	25.0	38.0	50.0
purchase_frequency_days	3900	89.13	119.04	7.0	14.0	30.0	90.0	365.0

Table 1: Summary statistics of numerical features (after cleaning)

- **Missing Data Handling:** Checked for null values and imputed the 37 missing values in `review_rating` using the median rating of each product category.
- **Duplicate Check:** Confirmed there were 0 duplicate records.
- **Column Standardization:** Renamed columns to **snake case** (lowercase with underscores) for readability.
- **Feature Engineering:**
- – Created **age_group** by splitting ages into four quartile-based groups: Young Adult, Adult, Middle-aged, Senior.
- – Created **purchase_frequency_days** by converting frequency labels (Weekly = 7, Monthly = 30, Quarterly = 90, Annually = 365, etc.) into days.
- **Data Consistency Check:** Verified that `discount_applied` and `promo_code_used` were identical; dropped `promo_code_used`.
- **Visual EDA:** Created a correlation heatmap, count plots, histograms, and box plots to understand distributions and relationships.

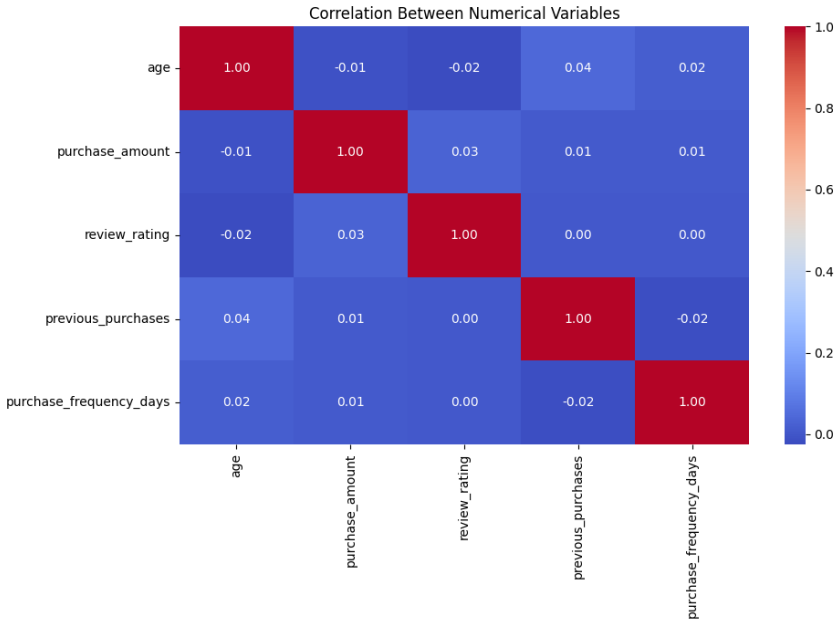


Figure 1: Correlation between numerical variables. All correlations are close to zero, so no strong linear relationships exist.

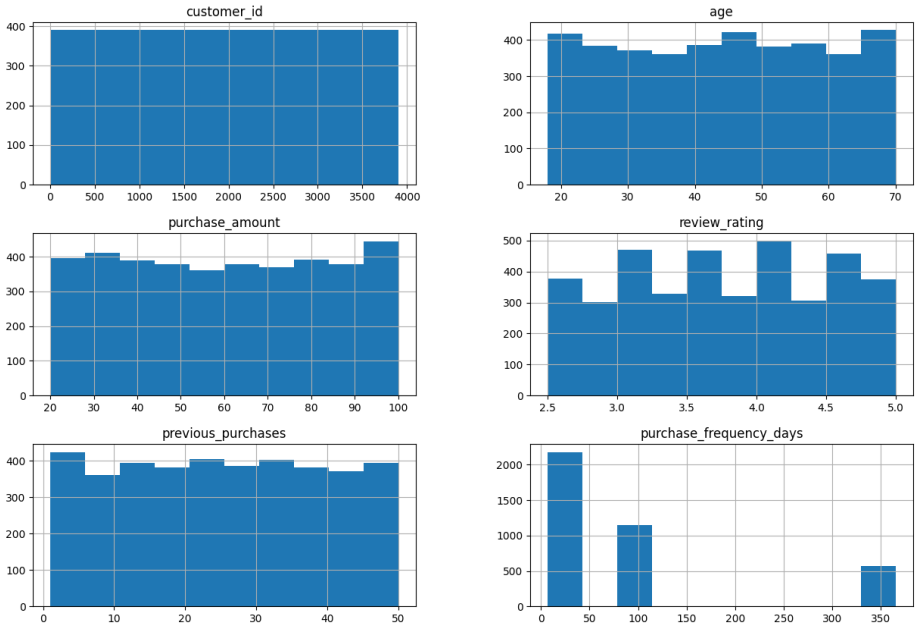


Figure 2: Distribution of numerical features. Age, purchase amount, and previous purchases are spread fairly evenly.

4. Data Analysis using SQL (Business Transactions)

We answered 32 business questions with SQL, grouped into six themes. Key results are shown below.

A. Sales and Revenue Analysis

1. Overall Sales Summary – Total revenue, average purchase, unique customers, and purchase range.

total revenue	avg purchase amount	unique customers	min amount	max amount
233081	59.76	3900	20	100

2. Revenue by Category – Identified which product categories generate the highest revenue.

category	total revenue	avg amount
Clothing	104264	60.03
Accessories	74200	59.84
Footwear	36093	60.26
Outerwear	18524	57.17

3. Top Products by Average Purchase Amount – Found the products with the highest average spend.

item purchased	total revenue	avg amount
T-shirt	9248	62.91
Boots	9018	62.63
Dress	10320	62.17
Shoes	9240	61.6
Shirt	10332	61.14
Scarf	9561	60.9

B. Customer Behavior Analysis

4. Revenue by Age Group – Calculated the total revenue contribution of each age group.

age group	total revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Revenue by Gender – Compared total and average revenue for male and female customers.

gender	revenue	avg amount
Female	75191	60.25
Male	157890	59.54

6. Subscribers vs. Non-Subscribers – Share of customers with a subscription.

subscription status	customers	percentage
No	2847	73.0
Yes	1053	27.0

7. Repeat Buyers – Customers with more than 20 previous purchases.

customers with over 20 previous purchases
2339

8. Purchase Frequency – Distribution of how often customers purchase.

frequency of purchases	customers
Every 3 Months	584
Annually	572
Quarterly	563
Monthly	553
Bi-Weekly	547
Fortnightly	542
Weekly	539

9. Category Preference by Age Group – Number of customers per category within each age group.

category	Young Adult	Adult	Middle-aged	Senior
Accessories	318	321	294	307
Clothing	479	411	426	421
Footwear	148	137	183	131
Outerwear	83	73	83	85

C. Discount and Promotion Analysis

10. Discount vs. No Discount – Compared the average purchase amount with and without a discount.

discount applied	customers	avg purchase amount
No	2223	60.13
Yes	1677	59.28

11. Discount Usage by Category – Categories with the highest share of discounted purchases.

category	total	discounted	discount pct
Outerwear	324	144	44.44
Accessories	1240	543	43.79
Footwear	599	259	43.24
Clothing	1737	731	42.08

12. Discount Usage by Age Group – Age groups that use discounts most often. Overall, 43% of purchases used a discount.

age group	total	discounted	discount pct
Senior	944	417	44.17
Adult	942	406	43.1
Young Adult	1028	443	43.09
Middle-aged	986	411	41.68

13. Subscribers vs. Discount Usage – Checked whether subscribers are more likely to use discounts.

subscription status	total	discounted	discount pct
Yes	1053	1053	100.0
No	2847	624	21.92

D. Shipping and Payment Analysis

14. Shipping Methods – Popularity and average spend by shipping type.

shipping type	orders	total amount	avg amount
Free Shipping	675	40777	60.41
Standard	654	38233	58.46
Store Pickup	650	38931	59.89
Next Day Air	648	37993	58.63
Express	646	39067	60.48
2-Day Shipping	627	38080	60.73

15. Payment Methods – Most used payment methods and average spend.

payment method	orders	total amount	avg amount
PayPal	677	40109	59.25
Credit Card	671	40310	60.07
Cash	670	40002	59.7
Debit Card	636	38742	60.92
Venmo	634	37374	58.95
Bank Transfer	612	36544	59.71

16. Shipping Preference of Subscribers – Most popular shipping methods among subscribers.

shipping type	subscriber orders
Store Pickup	190
Express	190
Free Shipping	177
Standard	175
Next Day Air	168
2-Day Shipping	153

E. Customer Satisfaction Analysis

17. Rating by Category – The overall average review rating is 3.75 out of 5.

category	avg rating	reviews
Footwear	3.79	599
Accessories	3.77	1240
Outerwear	3.75	324
Clothing	3.72	1737

18. Products with Ratings Below 3.5 – Top 8 products by number of low-rated reviews.

item purchased	low rated reviews	avg low rating
Shirt	77	2.96
Blouse	75	3.0
Pants	70	2.99
Jewelry	67	3.0
Scarf	65	2.98
Dress	65	3.05
Belt	65	3.0
Hoodie	64	2.99

19. Purchase Frequency vs. Rating – Checked whether frequent buyers give higher ratings.

frequency of purchases	avg rating
Monthly	3.77
Every 3 Months	3.77
Weekly	3.76
Fortnightly	3.76
Annually	3.76
Quarterly	3.73
Bi-Weekly	3.71

F. Product and Seasonal Insights

20. Most Popular Products by Season – Top 3 products in each season.

season	rank	item purchased	orders
Fall	1	Jacket	54
Fall	2	Hat	50
Fall	3	Handbag	48
Spring	1	Sweater	52
Spring	2	Shorts	47
Spring	3	Blouse	46
Summer	1	Pants	50
Summer	2	Dress	47
Summer	3	Jewelry	47
Winter	1	Sunglasses	52
Winter	2	Pants	51
Winter	3	Shirt	50

21. Most Purchased Colors – Top 5 colors.

color	orders
Olive	177
Yellow	174
Silver	173
Teal	172
Green	169

22. Most Purchased Sizes – Distribution of purchases by size.

size	orders
M	1755
L	1053
S	663
XL	429

23. Top Locations by Revenue – The 10 states with the highest total purchase amount.

location	total amount
Montana	5784
Illinois	5617
California	5605
Idaho	5587
Nevada	5514
Alabama	5261
New York	5257
North Dakota	5220
West Virginia	5174
Nebraska	5172

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually. It includes KPI cards (total customers, total revenue, average purchase, average review, categories, and items), slicers for subscription status, gender, category, shipping type, and season, and charts for revenue by category, age group, gender, season, and a seasonal heatmap by category.

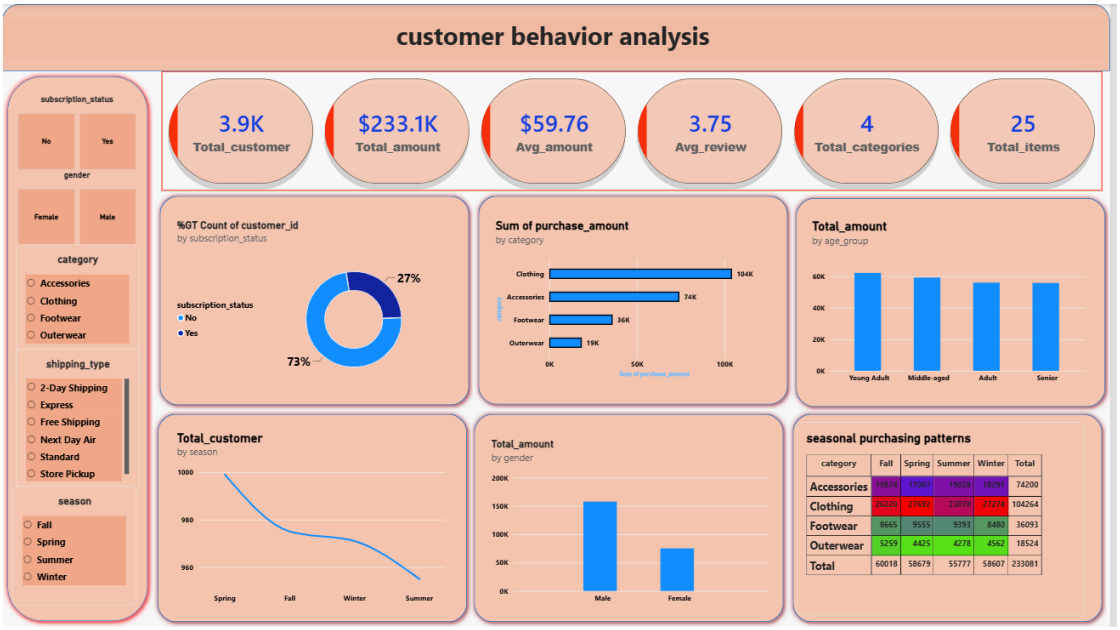


Figure 3: Customer Behavior Analysis dashboard in Power BI

6. Key Insights

- **Revenue:** Total revenue is **$233,081** with an average purchase of **$59.76** across 3,900 customers.

- **Categories:** Clothing leads with $104,264 (45%), followed by Accessories at $74,200. Outerwear is the smallest at $18,524.
- **Age groups:** Young Adults contribute the most revenue ($62,143), but all groups are within a narrow range.
- **Gender:** Male customers generate $157,890 versus $75,191 for female customers, even though average spend per purchase is nearly the same (about $60).
- **Subscriptions:** Only 27% of customers are subscribers. Every subscriber purchase used a discount, compared with 22% for non-subscribers.
- **Discounts:** 43% of purchases used a discount, and average spend is about the same with or without one ($59.28 vs. $60.13).
- **Shipping and payment:** Orders are spread evenly across shipping types and payment methods. Free Shipping and PayPal are the most used.
- **Satisfaction:** The average rating is 3.75 / 5. Footwear and Accessories score slightly higher than Clothing.
- **Locations:** Montana, Illinois, and California have the highest revenue.

7. Business Recommendations

- **Boost Subscriptions** – Only 27% of customers subscribe. Promote exclusive benefits to grow this segment.
- **Review Discount Policy** – Discounts do not raise average spend, so balance sales boosts with margin control. Since all subscriber purchases are discounted, consider tiered perks.
- **Grow Outerwear and Footwear** – These categories have lower revenue but good ratings, so they have room to grow with targeted campaigns.
- **Engage Female Customers** – Female customers spend as much per purchase but account for less total revenue, which points to an opportunity to reach more of them.
- **Customer Loyalty Programs** – Reward repeat buyers (2,339 customers have over 20 previous purchases).
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns, and address items with many low ratings.
- **Targeted Marketing** – Focus on high-revenue states and age groups, and use seasonal patterns to plan inventory.

8. Conclusion

This project shows how Python, SQL, and Power BI work together in an analytics workflow. Cleaning and feature engineering prepared the data, SQL answered targeted business questions, and the dashboard presented the results in a form decision-makers can explore. The analysis shows a balanced customer base where spending is driven more by category and subscription behavior than by demographics, and it points to clear opportunities in subscriptions, discount strategy, and category growth.